KONSTANTINOS KYRTSONIS

Results-driven professional with expertise in data pipeline development, model training and optimization. Experienced in end-to-end LLM app development and agent frameworks, including orchestration & APIs.
Strong managing skills, and outstanding communication capacity, with a proven track record and focus on delivering impactful solutions.

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| Born: 27 September 1990 in Athens, Greece| Permit: B

Professional Summary

CORE COMPETENCIES
Data Science | Deep Learning | Machine Learning | Automation | Innovative Thinking | Strategic Planning | Analytical Skills | Leadership Skills

Technical Skills: Python | Spark | LLM's | OpenAI API | PyTorch | Tensorflow | PowerBI | 3D (Blender, Revit)
Microsoft Office (Word, Excel, PowerPoint)

Language Skills: Greek – Native | **English** – C1 | **German** – C1

PROFESSIONAL EXPERIENCE
DATA SCIENTIST
Amberg Loglay AG | Zurich, Switzerland **02/2023 - Present**

- Building objects classification pipeline: Designed and implemented a classification system for processing building object data, through word embedding and multiclass classification system utilizing **OpenAI API**, **Google Gemini API** LLM's with Few-Shot prompting and alongside additional embedding technologies, Hugging Face Transformers (Sentence-Transformers), spaCy for text preprocessing to assign elements to building objects classification Catalogs, and to generate product descriptions reports and recommendations to any given new building dataset.

- Developing a multi-modal sensor framework for construction monitoring and activity recognition—integrating audio streams, and Bluetooth-beacon data to infer which working activity (which discipline is happening at any given time) which drilled-down task is happening in the construction site, delivery events, to compare real-time metrics with planned logistics schedules, and provide insights on optimization for scheduling and space management.The project was part of a Research Thesis supervised by University of Edinburgh and Frauenhofer Institute

- Developed and implemented a data analytics pipeline using **Python** to automate data ingestion, cleaning, transformation, and validation processes. Designed scalable solutions leveraging **Pandas**, **NumPy**, and for handling large and complex datasets from Building Information Modelling (BIM). Utilized **Streamlit** for visualizing the reports. Developed enhanced system performance by integrating parallel processing of building data and schedules, using **ETL** workflows, reducing processing manual time significantly.

- Application Development for 3D Space Recognition: Creation of a machine learning model to identify and categorize different spatial patterns in 3D models.

- 3D Reconstruction Application: Contributed with the R&D development of an application that uses computer vision to convert automatically 2D building floorplans into detailed, 3D models with **Blender** and **Python**. The app contributed to automate 50% of the old workflow to create the 3D Models and reduced significantly the manual working time.

3D MODEL ENGINEER - DEVELOPER **06/2021 – 10/2022**

EBP Schweiz AG | Zurich, Switzerland

- Concrete Crack Segmentation application development, using **CNN Deepcrack**, that depends on Deep Hierarchical Feature Learning Architecture) for Crack Segmentation on Structures like building walls, along with detecting various other characteristics of the cracks. Used: **Python**, **Pytorch**.
- Fulfill primary tasks by creating apps for 3D Models such as Autodesk Civil 3D API Addins and Autodesk Revit API Addins for the computational geometry manipulation with Python, providing sophisticated ways to create 3D models of real-world structures and buildings.
- Automatic classification of 3D Model elements to enable engineers to save time for modeling, increasing the company's productivity. Used point cloud types and typical 3D Models (IFC files)
- Streamline procurement and delivery schedules by creating BIM data dashboards with **PowerBI**, bringing improvements in the project control department through improved quality and cost avoidance. Highlights: 2D to 3D BIM Model tool creation.

DATA SCIENCE INTERN **08/2020 – 09/2020**

Fluence AG (Former Nispera AG), Data Science (Internal Project) | Zurich, Switzerland

- Aided the data analysts and other professionals in conducting complex statistical calculations by developing a software tool with Python, enabling the company to drive innovative data analytics solutions for a solar panel plant energy output optimization.

- Utilized analytical, statistical, and programming skills by collecting, analyzing, and interpreting data sets to use the information to develop data-driven solutions to difficult business challenges.

ENGINEERING SOFTWARE TECHNICIAN **12/2020 - 05/2021**

Office Haratori GmbH | Zurich, Switzerland

- Collaborated with architects and craftsmen to transform blueprints/technical drawings into 2D/3D digital models, delivering top-quality projects.
- Debugged software related problems for technical drawings that lead to higher-quality designs and faster project completion of the engineering work.
- Automated various AutoCAD pipelines, enabling a quicker planning phase and increased productivity.

GIS DATA ANALYST **10/2019 - 04/2020**

National Cadaster and Mapping Agency | Chalkis, Greece

- Used and analyzed property assets using the internal analyses tools as well as write technical reports & analysis with PowerBI, ensuring to meet high-quality standards set by the company.
- Developed and created GIS maps utilizing data coming from the asset databases with focus on price, location, population statistics

3D BIM MODEL ENGINEER - DEVELOPER **10/2018 – 03/2020**
Angelou – Pappas Architects | Chalkis, Greece

- Carried out vital roles by creating technical drawings and three-dimensional building designs, allowing efficient communication among engineers and can be kept as a record of the planning process.
- Achieved higher productivity by automating the plan creation process, using the AutoCAD API and Revit API, writing scripts with the C# language, and saving lots of hours of manual work.

INTERNSHIP IN CIVIL ENGINEERING **06/2018 - 09/2018**
Eratosthenes Corp. | Chalkis, Greece

- Demonstrated excellent collaboration skills by working with other Engineers to coordinate large-scale civil engineering projects.
- Accomplished the internship deliverables by analyzing project tasks and preparing and submitting progress reports to the lead designer or project managers, exceeding the overall expectations.
- Executed the construction of road design and evaluation of materials to ensure the implementation of quality standards, with progress reporting to the project leader.

EDUCATION

- **MSC IN DATA SCIENCE TECHNOLOGY AND INNOVATION,** University of Edinburgh, Edinburgh, United Kingdom

Research Project: Sensor-based construction monitoring and activity recognition—designing an end-to-end framework that fuses audio streams, and Bluetooth–beacon data to infer activity intensity, track material handling events, and align real-time site metrics with planned logistics schedules.

- **DATA SCIENCE BOOTCAMP,** Constructor University (Former Propulsion Academy AG), Zurich, Switzerland |

04/2020 – 08/2020

Project: Crack Detection – Crack detection for concrete structures

- **CIVIL ENGINEERING (BSc + Meng),** Democritus University of Thrace, Xanthi, Greece | **10/2013 – 05/2019**

Master thesis: Experimental Study on the Seismic Behavior Response of Concrete Building Elements presented on

- **INFORMATICS AND TELECOMMUNICATIONS CERTIFICATION:** University of Athens, Greece | **10/2008 – 10/2011**

Key Courses Completed:

Data Structures, Database Systems, Programming in C